

Measuring the Charge-diffusion Kernel of a Back-illuminated CCD in Orbit from Glancing Particle Tracks

Roy T. Smart^{a,*}, Charles C. Kankelborg^a

^aMontana State University, Department of Physics, P.O. Box 173840, Bozeman, MT 59717, USA

Abstract.

The lateral diffusion of photogenerated charge in the undepleted layer of a back-illuminated silicon sensor sets its point spread function and the statistics of its noise in the ultraviolet, but it is rarely measured on a flight sensor. We show that glancing particle tracks, which cross the full thickness of the sensor over a length of tens of pixels, sample the diffusion width at every depth in a single exposure, and we use them to measure the depth-dependent charge-diffusion kernel of the three CCDs of the Interface Region Imaging Spectrograph in orbit, from 10044 tracks found in level-1 images taken mostly inside the South Atlantic Anomaly. The width of the charge cloud at the back surface, the depth at which the field-free layer ends, the spread acquired drifting across the depletion region, and the probability that two electrons deposited at the same depth are collected in the same pixel are measured directly and compared with a field-free diffusion model. On the Slit-Jaw Imager the same-pixel probability at the back surface is 0.34 ± 0.00 , against 0.34 from the model, and the tracks require a spread of $0.80 \mu\text{m}$ inside the depletion region, which the field-free model neglects. The method needs no laboratory access and applies to any back-illuminated sensor in orbit.

Keywords: charge-coupled devices, charge diffusion, point spread function, ultraviolet detectors, particle tracks, IRIS.

*Roy T. Smart, roytsmart@gmail.com

1 Introduction

A back-illuminated silicon charge-coupled device (CCD) absorbs ultraviolet (UV) light within a fraction of a micron of its back surface, in a layer where there is no electric field to sweep the photogenerated charge toward the gates. The charge diffuses until it reaches the edge of the depletion region, and the resulting charge cloud, which can be several microns wide, is shared among neighboring pixels. This sharing broadens the point spread function (PSF) of the sensor¹ and, because the electrons liberated by one photon are shared together, it changes the statistics of the image noise, entering the variance-to-mean ratio (VMR) of a photon-transfer curve through the probability that two electrons from the same photon land in the same pixel.²

The width of the charge cloud is set by the thickness of the undepleted layer, which depends on the resistivity of the wafer and the bias applied to the device, both of which vary between nominally identical sensors and are rarely known for a sensor in orbit. Laboratory measurements use light of varying penetration depth^{3,4} or the split events of soft X-rays,⁵ and neither is available once the instrument has launched.

Energetic particles offer a probe that arrives for free. A particle crossing the sensor at glancing incidence deposits charge along a track tens of pixels long, and because it enters at the back surface and exits at the gates, the fractional position along the track is the fractional depth of the charge. Cosmic-ray muons have been used in this way to measure the growth of charge diffusion with drift distance in thick, fully depleted CCDs,⁶ where the cloud widens with depth. In a thin sensor with an undepleted back surface the dependence is reversed, the cloud is widest at the back surface and collapses to a pixel-sharp line in the depletion region, and the same tracks measure the field-free kernel that governs UV imaging.

Here we apply this method to the three CCDs of the Interface Region Imaging Spectrograph (IRIS),⁷ whose far ultraviolet (FUV) and near-UV photon-transfer curves⁸ motivated the question. Section 2 describes the tracks, the model of their cross-section, and the fit; Section 3 presents the kernel of each sensor; and Section 4 compares it with the field-free model and discusses the systematics, including the charge density of the tracks.

2 Method

2.1 Data

IRIS⁷ carries four CCDs of the same design: two behind the FUV channel of the spectrograph (FUV1 and FUV2, which share one level-1 image), one behind its near ultraviolet (NUV) channel, and one behind the Slit-Jaw Imager (SJI). They are back-illuminated, 14 μm thick, with 13 μm pixels, and their level-1 images are the raw frames with only the dark and the readout corrections applied, so a particle hit appears exactly as the camera recorded it. We searched 23792 such images, 8260 of them taken inside the South Atlantic Anomaly (SAA), where the flux of trapped protons at the altitude of IRIS is greatest,^{9,10} drawn from the campaigns of Table 1. The campaigns added for this work were chosen among those exposing for 8 s, since a longer exposure crowds the frame inside SAA with hits that spoil the tracks around them (the campaigns of the original measurement that pass through SAA expose for 15 s, and yielded per frame half as many of the tracks kept below), and among those whose spectrograph readout window lies mostly beyond the limb, as the pointing recorded in each frame’s header places it. Figure 1 shows a spectrograph image and a slit-jaw image taken seconds apart during one pass through SAA: the particle hits are dense, and a handful of them are the long glancing tracks we want. Tracks were sought only where the pointing recorded in each frame’s header places the pixel at least 15 arcseconds above the photospheric limb, so that no track lies on the disk or in the chromosphere and spicules seen just above the limb.

2.2 Finding tracks

The background of each frame was estimated as the trimmed mean of the frames in the same observing sequence and subtracted, and the read noise was estimated from the median absolute deviation of the residual. Pixels more than five times the read noise above the background were labeled, and a straight line was fit by least squares to the charge-weighted centroid of each row of every connected group of labeled pixels. We kept the groups spanning at least twelve rows (or columns) with a slope below 0.35 pixels per row, so that each track is nearly aligned with the pixel grid and its cross-section is sampled once per row without resampling the pixels, and extracted a seven-pixel-wide cutout centered on the integer part of the fitted line in every row. This yielded 10044 tracks. The slope cut is not as costly as it sounds: Figure 2 shows that the trapped protons arrive strongly aligned with the slit axis, so most of them pass it.

2.3 Model

Figure 3 shows the principle. A particle that enters the back surface and exits the front surface spans the full thickness of the sensor, so the fractional depth of slice i of an N -slice track is $t = (i + 1/2)/N$, up to the orientation of the track, and the charge deposited in each slice diffuses

Table 1 The observing campaigns searched for tracks, with the cameras kept from each (F for the far-ultraviolet spectrograph, N for the near-ultraviolet one, S for the slit-jaw imager), the spacecraft roll, the exposure time, and the number of frames searched, with the number taken inside SAA in parentheses. The tracks of every campaign but the two marked †, whose frames were taken outside SAA, are left by trapped protons; those two hold cosmic rays. The core tracks are the flat tracks with $0.25 < t_c < 0.6$, over which the mean t_c with its standard error and the median $\sigma_{\max}$ with its interquartile range are taken.

date	cameras	roll	exp. (s)	frames	tracks	flat	core	t_c	$\sigma_{\max} (\mu\text{m})$
2014-03-12	FN	0°	15	888 (114)	608	171	140	0.418 ± 0.006	5.0 [4.5, 6.0]
2014-04-07†	FN	0°	4	1950 (0)	34	15	13	0.408 ± 0.015	5.5 [4.5, 6.0]
2018-03-16†	FN	0°	15	622 (34)	118	50	38	0.392 ± 0.009	5.0 [4.0, 5.0]
2018-05-04	FN	−90°	15	722 (273)	611	174	129	0.406 ± 0.006	5.5 [4.5, 6.0]
2018-05-04/06	S	−90°	15	1156 (469)	471	157	139	0.410 ± 0.006	5.5 [5.0, 6.0]
2014-08-20	FNS	−75°	8	782 (333)	636	328	292	0.405 ± 0.004	5.5 [5.0, 6.0]
2014-08-21	FNS	−75°	8	768 (318)	800	367	328	0.403 ± 0.004	5.2 [4.5, 6.0]
2014-08-23	FNS	90°	8	1214 (333)	800	471	417	0.409 ± 0.003	5.5 [5.0, 6.0]
2014-08-24	FNS	90°	8	806 (330)	942	473	436	0.408 ± 0.003	5.0 [4.5, 6.0]
2014-08-25	FNS	90°	8	778 (321)	1482	641	547	0.408 ± 0.003	5.5 [4.5, 6.0]
2014-11-02	FNS	0°	8	762 (300)	1153	463	388	0.399 ± 0.003	5.5 [5.0, 6.0]
2014-11-27	FNS	0°	8	970 (383)	144	9	9	0.406 ± 0.024	5.5 [4.5, 5.5]
2015-03-26	FNS	−90°	8	616 (210)	65	33	30	0.398 ± 0.011	5.5 [4.5, 6.0]
2015-09-20	FNS	0°	8	1039 (575)	505	245	221	0.406 ± 0.005	5.5 [5.0, 6.0]
2016-01-02	FNS	0°	8	3836 (1134)	542	218	183	0.406 ± 0.005	5.5 [4.5, 6.0]
2016-01-06	FN	0°	8	638 (184)	90	61	58	0.423 ± 0.009	5.0 [4.5, 5.5]
2016-05-16	FNS	0°	8	1174 (732)	511	243	208	0.397 ± 0.004	5.5 [5.0, 6.0]
2016-10-28	FN	0°	8	556 (234)	186	98	82	0.415 ± 0.007	5.0 [4.5, 5.5]
2016-12-13	FNS	0°	8	1737 (990)	123	67	65	0.389 ± 0.007	5.0 [4.5, 5.5]
2016-12-17	FNS	0°	8	1247 (192)	3	2	2	0.350 ± 0.035	5.2 [4.9, 5.6]
2016-12-25	FNS	0°	8	1531 (801)	220	101	95	0.404 ± 0.007	5.0 [4.5, 6.0]

from a known depth. We model the charge in every slice as spread uniformly along the fitted centerline and diffused as a Gaussian whose variance has two parts,

$$\sigma^2(t) = \sigma_{\max}^2 \left(1 - \frac{t}{t_c}\right) + \sigma_d^2 g(t). \quad (1)$$

The first part is the field-free layer. It applies only for $t < t_c$, where $t_c = z_f/D$ is the fractional thickness of the layer and $\sigma_{\max}$ is the width of the charge cloud at the back surface, and it is the depth dependence of a random walk to an absorbing plane at z_f , the form used to model the UV noise of these sensors in the companion article. The second part is the depletion region. Charge that reaches the edge of the depletion region, whether it was deposited there or diffused there from the field-free layer, still has to drift to the gates and spreads a little as it does so. σ_d is the spread acquired crossing the full depleted thickness, and $g(t) = \min[(1 - t)/(1 - t_c), 1]$ scales it with the distance drifted, so that it is one for all charge deposited in the field-free layer and falls linearly to zero at the gates. The field-free model of the companion article is the case $\sigma_d = 0$. We work in fractional depth because we observe the length of a track in slices rather than the thickness of the sensor in microns, and the uniform spread along the tilted centerline within a slice is approximated by adding $m^2/12$ to the variance, where m is the slope in pixels per row.

The three parameters are not fit alike. t_c and $\sigma_{\max}$ describe the field-free layer, whose thickness can vary across a wafer, and are fit to each track together with its orientation, by exhaustive search



Fig 1 Level-1 images taken seconds apart during a pass through SAA on 2018 May 4, with the spacecraft rolled by -90° so that the limb runs horizontally across the field. (a) The slit-jaw image at 2796 Å, whose lower half is off the limb. (b) The widest of the readout windows of the spectrograph image, on the FUV2 CCD. (c) One of the tracks in (a) at full resolution. The images are shown with an inverse hyperbolic sine stretch above the background, saturating at 400 data numbers, and the red boxes outline the tracks the finder kept from each frame. Most of the particle hits are the short streaks and dots that pepper both images; the tracks used here are the long, thin, nearly vertical ones.

on a grid of 21 values in each of $t_c \in [0, 1]$ and $\sigma_{\max} \in [0, 10] \mu\text{m}$. At each grid point the offset and tilt of the centerline are adjusted, by up to 0.6 pixels and 0.03 pixels per row, to minimize a robust misfit, $\sum \ln(1 + r^2/2)$, where r is the residual of the fraction of each slice's charge in each pixel in units of the read noise. This is the negative log-likelihood of a Cauchy distribution rather than a Gaussian, so that a single stray pixel, from a delta ray or a second hit inside the cutout, cannot dominate the fit. σ_d describes the drift field, which is a property of the CCD and its bias rather than of any one track, and a single track constrains it only weakly, so it is fit to each CCD as a whole. Every track is fit as above at each of 13 values of σ_d from 0 to 3 μm , the misfits of the flat tracks, defined next, are summed at each value, and the value that minimizes the sum is adopted for every track on that CCD. Which tracks are flat depends on their fits, so the selection and σ_d are iterated to consistency, which takes one or two rounds.

2.4 Selection

The useful tracks are those which cross the full thickness of the sensor at nearly constant energy loss. We kept only the tracks for which the fit constrains t_c to within 0.15 and improves on a model with no diffusion by at least ten units of misfit, for which the median charge per slice in the last third of the track is within 50% of that in the first third, since a rise in the deposited charge along the track is the Bragg peak of a particle that stopped inside the sensor and did not cross its full thickness, and for which the fit finds a depleted end, $t_c \leq 0.7$, since a particle that crossed the sensor is pixel-sharp where it left through the gates, whereas a feature that is wide from end to end is, on the slit-jaw imager, usually a spicule or other structure at the limb that the mask let through. We call these the flat tracks; 4387 tracks pass these cuts, 1016 of them on the SJI CCD, and Figure 8 shows the longest of them alongside a stopping track.



Fig 2 The azimuth in the detector frame of every elongated connected component above 5σ in the spectrograph frames, with a charge-weighted width under 3 pixels and an aspect ratio over 4, before any of the finder's cuts, so that the whole circle is populated. Tracks are undirected, so each is drawn at θ and $\theta + 180^\circ$, and the grey wedges are the windows about the row and column axes that the finder accepts. Left: the frames taken inside SAA at both spacecraft rolls. Right: quiet frames at the same rolls. The trapped protons arrive strongly aligned, mostly within the window along the slit, while the cosmic rays are isotropic to within counting noise; the preferred direction shifts and broadens when the spacecraft is rolled, so it is a property of the arrival direction as seen by the instrument rather than of the detector.

3 Results

3.1 The width against depth

Figure 4 shows the flat tracks stacked on their fitted centerlines and the diffusion width they imply at each depth, measured two ways. The first assumes no depth dependence at all: for each depth bin, the misfits of every slice in the bin are summed over a grid of trial widths and the width which minimizes the sum is taken, so the wedge shape emerges from the data rather than being imposed. The second is the average of the per-track fits of Equation 1. The two agree to within a few tenths of a micron on all three CCDs at every depth. Beyond t_c the model-free profile does not go to zero but keeps a floor of 0.5 to 1 μm across the depletion region, and the fits follow it: this floor is the σ_d term of Equation 1.

3.2 The spread inside the depletion region

Figure 5 shows how σ_d is determined. The misfit summed over the flat tracks on each CCD has a sharp minimum, at $\sigma_d = 0.80 \mu\text{m}$ on SJ1, $0.70 \mu\text{m}$ on FUV2, and $0.70 \mu\text{m}$ on FUV1, with the neighboring grid points a quarter micron away already disfavored by about a hundred units of misfit. The improvement over $\sigma_d = 0$ is three to four hundred units pooled but only two or three per

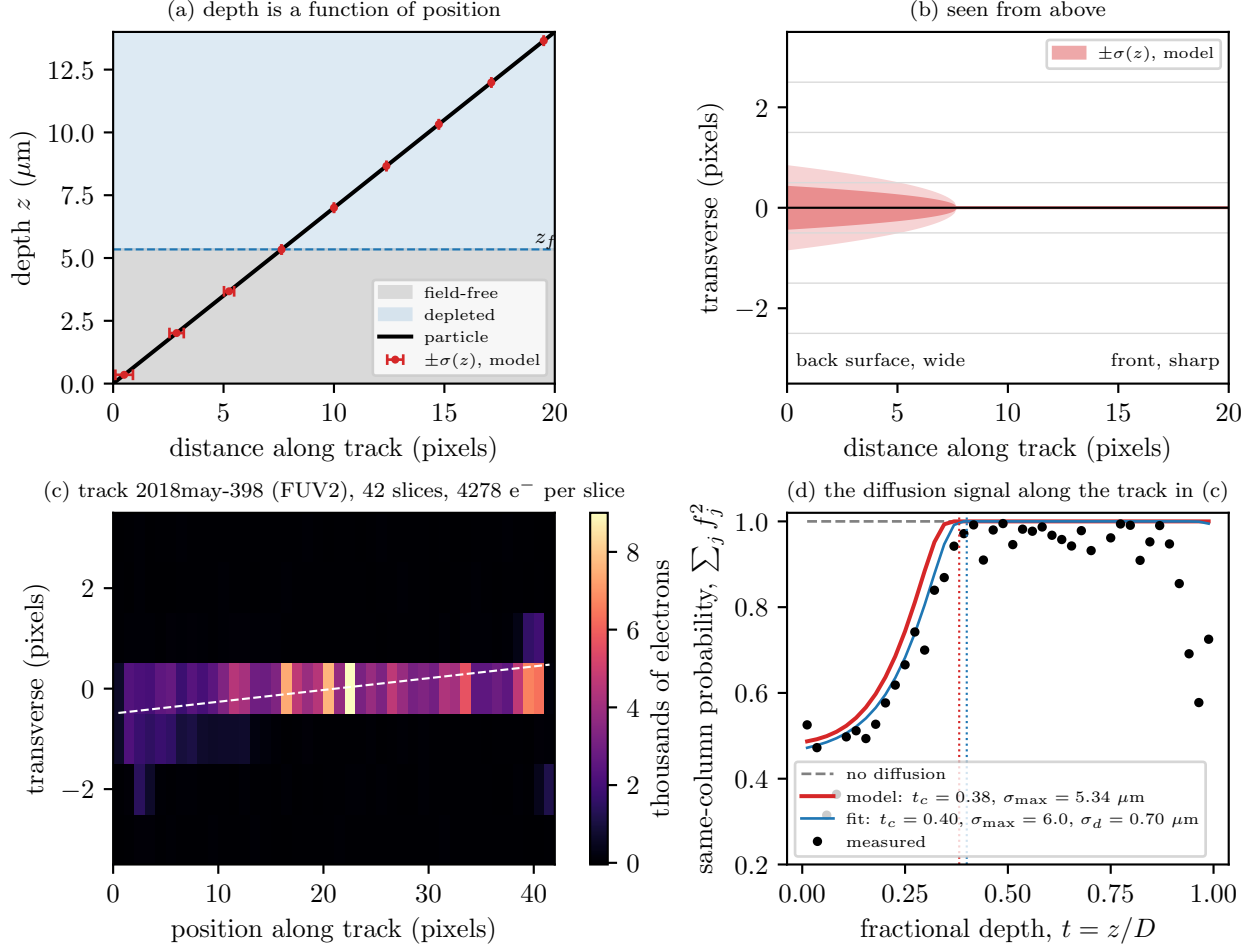


Fig 3 How a glancing track measures the diffusion width against depth. (a) A particle entering at the illuminated back surface ($z = 0$) and leaving at the gates ($z = D$) crosses the field-free layer first, where the charge it liberates diffuses until it reaches the depletion edge, so the lateral spread (red bars, to scale in pixels) is largest at the back surface and vanishes at $z = z_f$. (b) Seen from above, the track is a wedge: about 0.4 pixels wide at one end and pixel-sharp beyond t_c . (c) A proton track on the FUV2 CCD, in the coordinates of its parent image, with the fitted centerline dashed. (d) The probability that two electrons deposited in the same slice of that track are collected in the same column, $\sum_j f_j^2$ corrected for read noise, for every slice, against the field-free model, the best fit, and a model with no diffusion, all evaluated at the fitted centerline. The dip near $t = 0.2$ is where the centerline crosses a pixel boundary; the diffusion signal is the gap between the measured points and the no-diffusion curve.

track, and the value each track prefers on its own is spread over 0 to 2 μm , so the effect is modest for any one track and consistent across them, which is why σ_d is fit to each CCD as a whole. The resolution of the centerline matters here: a centerline located only to 0.1 pixel is uncertain by 1.3 μm , the size of the effect being sought, and with such a grid the minimum washes out entirely. σ_d trades against the field-free wedge: as it rises from zero to its adopted value, the median t_c of the flat tracks falls by 0.05 on every CCD and the median σ_{max} rises by half a micron on FUV2, since without it the wedge stretches to absorb the floor. Leaving σ_d out of the model would therefore bias both of the parameters the noise model needs, which is why it is part of the model rather than a correction applied afterwards.

3.3 The critical depth and back-surface width

Figure 6 shows the fitted t_c and $\sigma_{\max}$ of every flat track, and Table 1 their means in each campaign. The critical depth clusters between 0.35 and 0.50 on all three CCDs, around the 0.38 of the field-free model, and the two particle populations, the two spacecraft rolls, and the two cameras agree to within a few hundredths. The back-surface width does differ between chips: FUV2 and SJI cluster at 5.5 to 6.5 μm while FUV1 clusters at 3 to 5 μm , so the pooled scatter is mostly this chip-to-chip difference, which is expected from the resistivity of the wafer and the applied bias and is one reason a sensor's kernel is worth measuring rather than assuming.

3.4 The same-pixel probability

For the noise model the quantity that matters is not the width but the probability that two electrons deposited at the same depth are collected in the same pixel. For a slice with pixel fractions f_j that is $p = \sum_j f_j^2$, corrected for the read-noise contribution $\sum_j \epsilon_j^2$, and it can be read directly off each slice with no model of the shape of the kernel. Since the kernel is separable, the probability that two electrons are collected in the same pixel is $\mathcal{P} = p^2$. Figure 7 shows p against depth on each CCD, and Table 2 lists it for the slices within $D/10$ of the back surface, alongside the prediction of the field-free model evaluated at the fitted centerline of every track. On the SJI CCD the back surface gives $\mathcal{P} = 0.34 \pm 0.00$ against 0.34 from the model, and FUV2 agrees as well, while FUV1 spreads its charge over fewer pixels. Beyond t_c the measured probability settles at about 0.85 rather than the 0.95 of a perfectly sharp cloud on all three CCDs, which the per-track fits reproduce through σ_d and the field-free model does not.

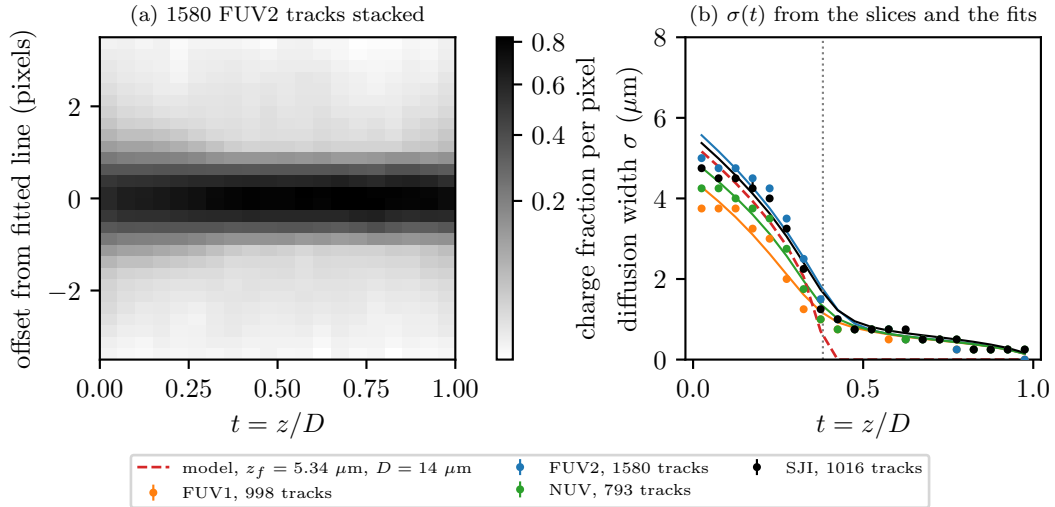


Fig 4 (a) Every slice of every flat track on the FUV2 CCD placed at its fractional depth and aligned on its fitted centerline, with each pixel's charge spread uniformly over its own width. The wedge is subtle: the wings at $t < 0.4$ carry a few percent of the charge. (b) Points: the diffusion width in each depth bin on each CCD, found by summing the misfit of every slice in the bin over a grid of trial widths and minimizing, with no parametric model of the depth dependence; the error bars span the widths within two units of misfit of the minimum. Lines: the average of the per-track fits of Equation 1 on the same CCD, which reproduces the model-free profile at every depth, including the floor of 0.5 to 1 μm beyond t_c (dotted) that the σ_d term supplies and the dashed field-free model sets to zero.

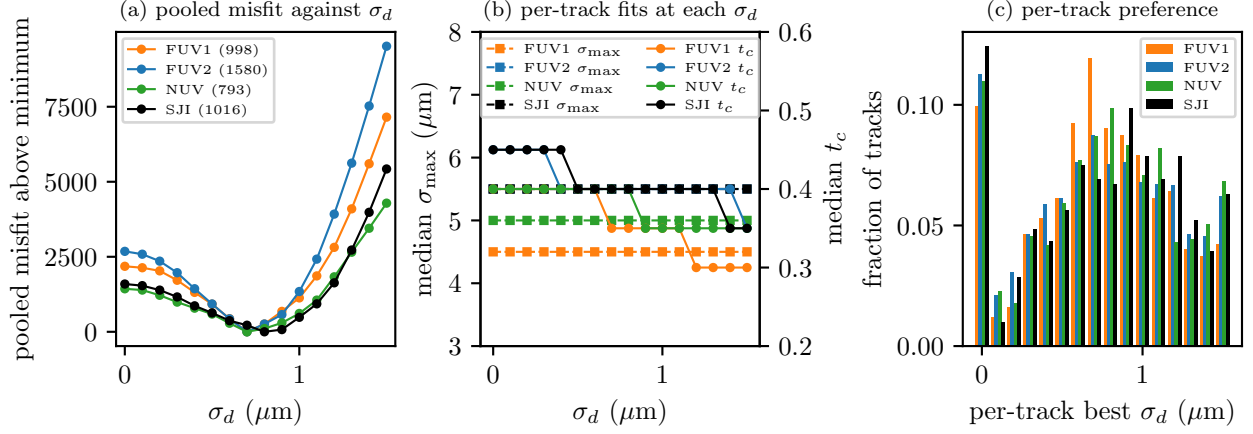


Fig 5 The spread inside the depletion region, σ_d in Equation 1, which is shared by every track on a CCD. (a) The misfit summed over the flat tracks on each CCD at each σ_d , with each track refit in t_c , σ_{max} , orientation, and centerline, shown relative to its minimum, which lies at $\sigma_d = 0.70$ μm on FUV1, 0.70 μm on FUV2, 0.70 μm on NUV, and 0.80 μm on SJI. (b) The median t_c and σ_{max} of the same tracks at each σ_d : the spread inside the depletion region trades against the field-free wedge, lowering t_c and raising σ_{max} . (c) The σ_d each flat track prefers on its own. The preference of any one track is weak, spread over the whole grid, and it is only in the sum that the minimum is sharp.

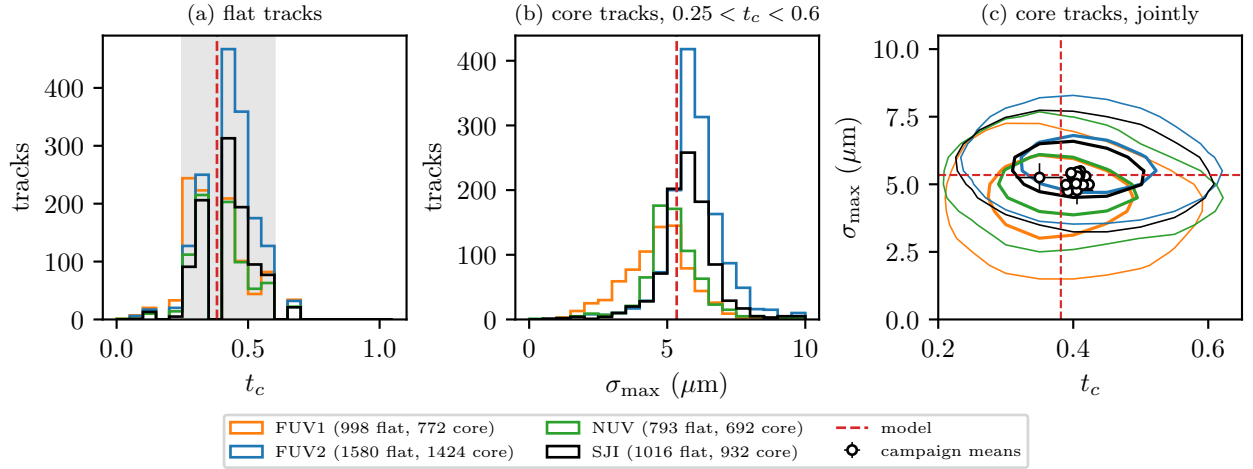


Fig 6 The fitted parameters. (a) The critical depth of every flat track on each CCD, with the field-free model dashed. The tails below 0.25 and above 0.6 are tracks the model describes poorly, sharp or diffuse along most of their length, and they are excluded from the shaded core used elsewhere. (b) The back-surface width of the core tracks. (c) The joint distribution of the core tracks on each CCD, as the contours enclosing half (thick) and nine tenths (thin) of them, on the grid the fits are searched over (0.05 in t_c and 0.5 μm in σ_{max}) smoothed by one cell, with the field-free model at the crossing of the dashed lines and the mean of each campaign of Table 1 with its standard error as an open circle. The campaigns, spanning two particle populations, four spacecraft rolls, and three cameras, agree to within a few hundredths in t_c , all within 0.04 of the model, and the contours run wider in σ_{max} than in t_c , since a single track constrains its back-surface width less well than the depth at which it sharpens.

4 Discussion

At the back surface, where UV photons are absorbed, the tracks confirm the field-free model on the SJI and FUV2 CCDs: the same-pixel probability of 0.34 ± 0.00 on SJI matches the 0.34

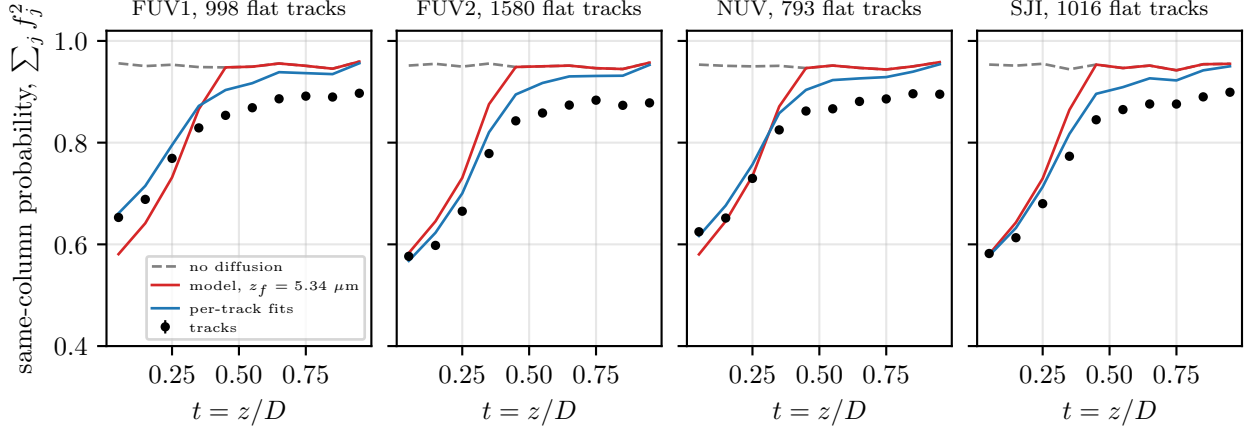


Fig 7 The probability that two electrons deposited in the same slice are collected in the same column, averaged over the flat tracks on each CCD in bins of fractional depth, with the standard error of each mean. The red line is the field-free model, Equation 1 with $t_c = 0.38$ and $\sigma_{\max} = 5.34 \mu\text{m}$, evaluated at the fitted centerline of each track and averaged in the same bins; the blue line is the same for each track’s own fit; and the dashed line is the same with no charge diffusion, which is below one only where a centerline runs near a pixel boundary. The back surface agrees with the model on the FUV2 and SJI CCDs, while beyond t_c the tracks settle below the no-diffusion curve on all three.

Table 2 The charge diffusion measured from glancing particle tracks on the IRIS CCDs. The number of frames is the number of level-1 images searched, with the number taken inside SAA in parentheses. The flat tracks are those which constrain t_c to within 0.15 and show no Bragg rise along their length. σ_d is the spread inside the depletion region fit to each CCD as a whole, and t_c and $\sigma_{\max}$ are the medians of the per-track fits at that σ_d , with the interquartile range in brackets. p is the probability that two electrons deposited within $D/10$ of the back surface are collected in the same column, $\mathcal{P} = p^2$ is the corresponding probability that they are collected in the same pixel, and the last column is $\mathcal{P}$ predicted by the field-free model for the same tracks.

CCD	frames	tracks	flat	σ_d (μm)	t_c	$\sigma_{\max}$ (μm)	p	$\mathcal{P}$	$\mathcal{P}_{\text{model}}$
FUV1	8471 (2755)	2369	998	0.70	0.35 [0.30, 0.45]	4.5 [3.5, 5.0]	0.653 ± 0.004	0.426 ± 0.006	0.338
FUV2	8471 (2755)	3360	1580	0.70	0.40 [0.35, 0.45]	5.5 [5.0, 6.1]	0.576 ± 0.003	0.332 ± 0.003	0.340
NUV	8474 (2755)	2036	793	0.70	0.40 [0.35, 0.45]	5.0 [4.5, 5.5]	0.625 ± 0.004	0.390 ± 0.005	0.337
SJI	6847 (2750)	2279	1016	0.80	0.40 [0.35, 0.45]	5.5 [5.0, 6.0]	0.582 ± 0.003	0.339 ± 0.004	0.337

predicted from the depletion thickness of the model, which is the quantity the companion noise model needs. Deeper in the sensor the tracks see a spread of about a micron, σ_d , that the field-free model neglects. It is about what the drift-time diffusion across the depleted thickness plus the micron-scale ionization column of the track itself should give, and it is small compared with the back-surface spread, so it has little effect on UV imaging. It cannot be left out of the fit, though: without it the wedge stretches to absorb the floor and t_c comes out too large, which is why Equation 1 carries it from the start. With it, the fitted t_c of 0.400 ± 0.003 on SJI is to be compared with the model’s 0.38.

The FUV1 CCD is narrower than the other two, at the back surface and in the fitted $\sigma_{\max}$ alike. We have no explanation beyond chip-to-chip variation of the field-free thickness, which depends on the resistivity of the wafer and the applied bias; the same measurement on the Atmospheric Imaging Assembly (AIA) CCDs, which are of the same design, would show whether such variation is common.

Two systematics deserve comment. First, the tracks are left by protons that deposit thousands

of electrons per row, whereas a UV photon liberates one to three. In thick, fully depleted sensors, slow protons leave tracks visibly widened by the mutual repulsion of that charge,¹¹ and if the same occurred in the field-free layer here the tracks would overestimate the single-photon kernel. The tracks themselves can test this, by comparing the kernels of faint and bright tracks, and that comparison is the first thing to add. Second, the depth assignment assumes every flat track crosses the full thickness of the sensor; a track that entered or left through the edge of the field-free layer, or that was clipped by the Bragg cut at one end, would have its depths compressed. The agreement between the model-free and parametric profiles, and between campaigns, suggests such tracks are rare, but a validation on synthetic tracks injected into real frames is the way to bound it.

5 Conclusion

Glancing particle tracks measure the depth-dependent charge-diffusion kernel of a back-illuminated CCD in orbit, with no laboratory access, and on the IRIS CCDs they confirm the field-free model at the back surface and measure the spread of about a micron acquired inside the depletion region that the model neglects.

Appendix A: The tracks as the finder cut them out

The cutouts distributed with this article are seven pixels wide and follow the fitted centerline, so they show the tracks after the finder has done its work. Figure 8 shows the longest of the flat tracks and one track that the Bragg cut rejects. Sub-pixel structure is invisible at this scale, which is why the wedge signature has to be extracted by fitting.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper. The analysis code and the text of this article were drafted with the assistance of Claude, a large language model, under the direction and review of the authors.

Code, Data, and Materials Availability

This article is executable:¹² every figure, table, and number quoted in the text is computed from the tracks when the article is built. The code, the track cutouts, and the fits are available at <https://github.com/roytsmart/ccd-diffusion-paper>, and the level-1 IRIS images they were extracted from are available from the IRIS data archive.

Acknowledgments

This work was supported by NASA through the IRIS mission.

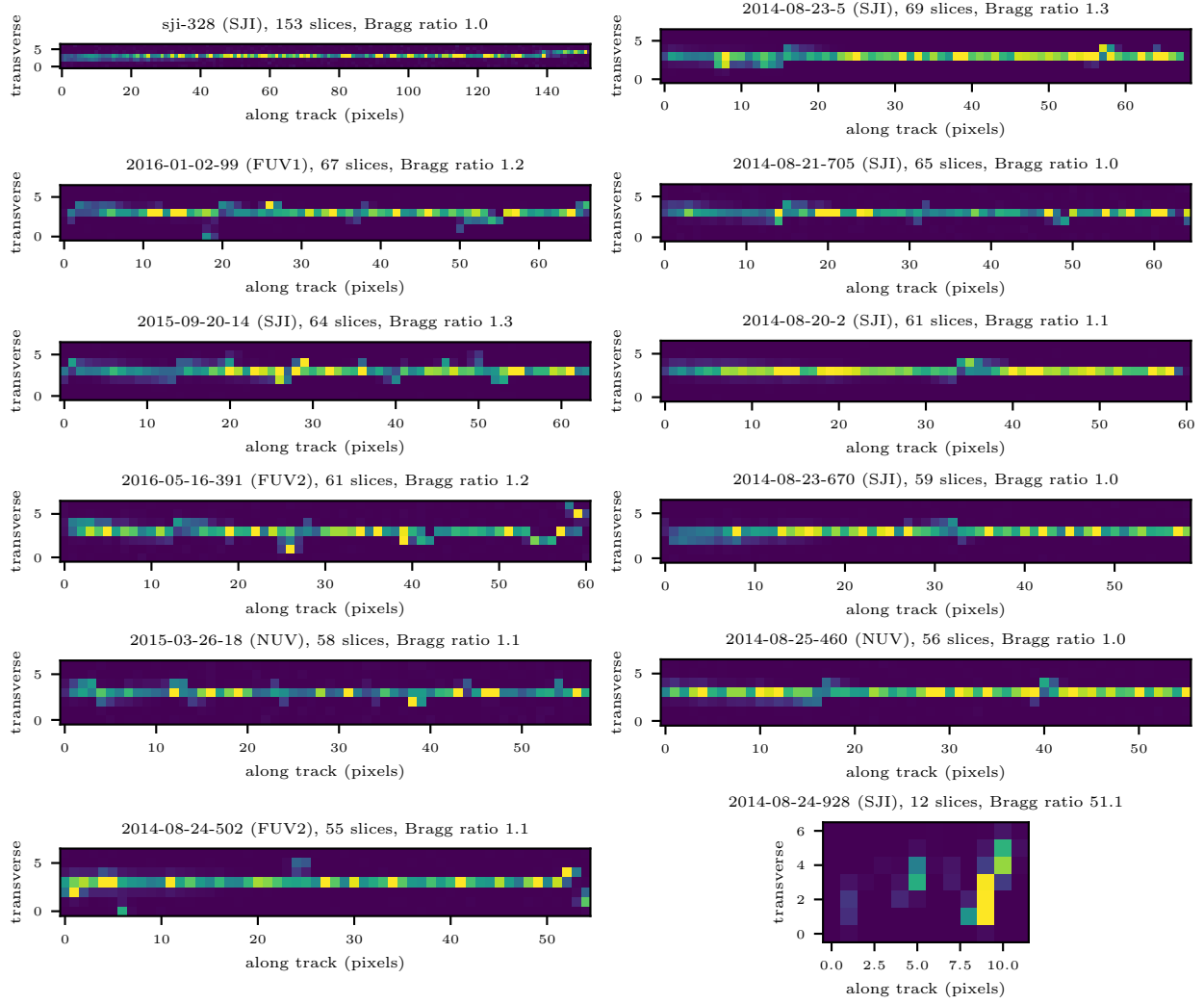


Fig 8 The eleven longest flat tracks and, in the last panel, the track with the strongest Bragg rise, as the finder cut them out: seven pixels wide, re-centered on the integer part of the fitted centerline in every slice, and oriented with the back surface at the left. Each panel’s color scale is clipped at its 98th percentile so that a single bright pixel does not hide the track. The stopping track’s charge per slice climbs by a factor of twenty toward one end, and the selection of Section 2 rejects it.

References

- 1 G. G. Pavlov and J. A. Nousek, “Charge diffusion in CCD X-ray detectors,” *Nuclear Instruments and Methods in Physics Research A* **428**, 348–366 (1999).
- 2 J. R. Janesick, *Scientific charge-coupled devices* (2001).
- 3 R. A. Stern, L. Shing, P. R. Catura, M. D. Morrison, D. W. Duncan, J. R. Lemen, T. Eaton, P. J. Pool, R. Steward, D. M. Walton, and A. Smith, “Characterization of the flight CCD detectors for the GOES N and O solar x-ray imagers,” in *Telescopes and Instrumentation for Solar Astrophysics*, S. Fineschi and M. A. Gummin, Eds., *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series* **5171**, 77–88 (2004).
- 4 R. Widenhorn, A. Weber, M. M. Blouke, A. J. Bae, and E. Bodegom, “Charge diffusion in the field-free region of charge-coupled devices,” *Optical Engineering* **49**(4), 044401 (2010).

- 222 5 G. W. Fraser, A. F. Abbey, A. Holland, K. McCarthy, A. Owens, and A. Wells, “The X-ray
223 energy response of silicon Part A. Theory,” *Nuclear Instruments and Methods in Physics
224 Research A* **350**, 368–378 (1994).
- 225 6 M. Fisher-Levine and A. Nomerotski, “Characterising CCDs with cosmic rays,” *Journal of
226 Instrumentation* **10**(08), C08006 (2015).
- 227 7 B. De Pontieu, A. M. Title, J. R. Lemen, G. D. Kushner, D. J. Akin, B. Allard, T. Berger,
228 P. Boerner, M. Cheung, C. Chou, J. F. Drake, D. W. Duncan, S. Freeland, G. F. Heyman,
229 C. Hoffman, N. E. Hurlburt, R. W. Lindgren, D. Mathur, R. Rehse, D. Sabolish, R. Seguin,
230 C. J. Schrijver, T. D. Tarbell, J. P. Wülser, C. J. Wolfson, C. Yanari, J. Mudge, N. Nguyen-
231 Phuc, R. Timmons, R. van Bezooijen, I. Weingrod, R. Brookner, G. Butcher, B. Dougherty,
232 J. Eder, V. Knagenhjelm, S. Larsen, D. Mansir, L. Phan, P. Boyle, P. N. Cheimets, E. E.
233 DeLuca, L. Golub, R. Gates, E. Hertz, S. McKillop, S. Park, T. Perry, W. A. Podgorski,
234 K. Reeves, S. Saar, P. Testa, H. Tian, M. Weber, C. Dunn, S. Eccles, S. A. Jaeggli, C. C.
235 Kankelborg, K. Mashburn, N. Pust, L. Springer, R. Carvalho, L. Kleint, J. Marmie, E. Maz-
236 manian, T. M. D. Pereira, S. Sawyer, J. Strong, S. P. Worden, M. Carlsson, V. H. Hansteen,
237 J. Leenaarts, M. Wiesmann, J. Aloise, K. C. Chu, R. I. Bush, P. H. Scherrer, P. Brekke,
238 J. Martinez-Sykora, B. W. Lites, S. W. McIntosh, H. Uitenbroek, T. J. Okamoto, M. A.
239 Gummin, G. Auker, P. Jerram, P. Pool, and N. Waltham, “The Interface Region Imaging
240 Spectrograph (IRIS),” *Solar Physics* **289**, 2733–2779 (2014).
- 241 8 J. P. Wülser, S. Jaeggli, B. De Pontieu, T. Tarbell, P. Boerner, S. Freeland, W. Liu, R. Tim-
242 mons, S. Brannon, C. Kankelborg, C. Madsen, S. McKillop, J. Prchlik, S. Saar, N. Schanche,
243 P. Testa, P. Bryans, and M. Wiesmann, “Instrument Calibration of the Interface Region Imag-
244 ing Spectrograph (IRIS) Mission,” *Solar Physics* **293**, 149 (2018).
- 245 9 J. L. Barth, C. S. Dyer, and E. G. Stassinopoulos, “Space, atmospheric, and terrestrial radia-
246 tion environments,” *IEEE Transactions on Nuclear Science* **50**(3), 466–482 (2003).
- 247 10 O. Adriani, G. C. Barbarino, G. A. Bazilevskaya, *et al.*, “Trapped proton fluxes at low Earth
248 orbits measured by the PAMELA experiment,” *The Astrophysical Journal Letters* **799**(1), L4
249 (2015).
- 250 11 T. A. Grosson and A. Nomerotski, “Fat cosmic ray tracks in charge-coupled devices,” *Journal
251 of Astronomical Telescopes, Instruments, and Systems* **9**(2), 028006 (2023).
- 252 12 J. Lasser, “Creating an executable paper is a journey through Open Science,” *Communica-
253 tions Physics* **3**, 143 (2020).